

ASRAF SIDDIQUEE

🏠 South Bend, Indiana, USA

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Summary

Mechanical and robotics engineer with strong expertise in mechanical design, simulation, and experimental validation of robotic and electromechanical systems. Proficient in CAD-driven product development, nonlinear FEA, multiphysics simulation, and physics-informed modeling, with hands-on experience in embedded hardware prototyping, sensor integration, and additive manufacturing. Experienced in leading research-driven hardware projects across underwater robotics, soft robotics, and energy systems, from concept through testing and deployment.

Education

Ph.D. Candidate in Electrical Engineering [University of Notre Dame, Notre Dame, IN]	Expected Graduation: 2027
M.S. in Electrical Engineering [University of Notre Dame, Notre Dame, IN]	2023
Master of Engineering Management [Arkansas State University, Jonesboro, AR]	2022
M.S. in Mechanical Engineering [Arkansas State University, Jonesboro, AR]	2017
B.S. in Mechanical Engineering [Khulna University of Engineering & Technology, Bangladesh]	2013

Technical Skills

- **Mechanical Design & Product Development:** SolidWorks, Fusion 360 (Parametric & Generative), Siemens NX, AutoCAD, GD&T, Design for Manufacturing (DFM), Tolerance Analysis, Multi-Material Integration.
- **Rapid Prototyping & Fabrication:** 3D Printing (FDM/SLA), Laser Cutting, Resin Casting, Multimaterial Integration (Plastics, Elastomers, Fabrics, and Softgoods), Stitching, CNC Machining, Sheet Metal Fabrication, Stamping, Forging, and Sand Casting, Plastic Injection Molding.
- **Robotics & Mechatronics:** Soft Robotics, Underwater Robotics, Actuation Systems, Embedded Control, Sensor Integration, System-Level Hardware–Software Integration, Proteus, Altium, KiCad.
- **Simulation & Modeling:** Abaqus, COMSOL, ANSYS, Multiphysics Simulation.
- **Programming:** Python, MATLAB, Embedded C++, Arduino, ESP32, Raspberry Pi.

Industry Experience

AceOne Technologies

HARDWARE ENGINEER INTERN

Jonesboro, AR, USA

Jan. 2020 – Jul. 2021

- **Developed embedded C++/Arduino firmware** for real time multi sensor data acquisition.
- **Designed and validated PackRight device hardware**, including enclosure redesign and high accuracy temperature sensor integration.

A-State Innovate, The Garage

MECHANICAL ENGINEER (VOLUNTEER)

Jonesboro, AR, USA

May. 2019 – Jan. 2020

- Designed and fabricated **four pneumatic grippers** with custom mounting mechanisms and multi-material integration.
- Designed and fabricated mechanical prototypes including a **multi cylinder mechanism, helical bevel gear**, stair structure, and table structure using CAD, additive manufacturing, GD&T, and DFAM

Adroit Engineers

MECHANICAL DESIGN & PROTOTYPING ENGINEER

Dhaka, Bangladesh

Oct. 2013 – Jan. 2015

- **Led mechanical design, FEA, and embedded system development (PCB, C/C++)** for electromechanical medical devices, including automated wheelchairs and hospital beds.
- **Optimized structural design** of automated wheelchair systems, achieving **7% reduction in weight and cost**.
- **Led mechanical product development** from design through prototyping and validation; developed an automated wheelchair with gearbox and linear-actuator systems and designed a gearbox mechanism for hospital-bed adjustment.

Research Experience

miNiRo Lab, University of Notre Dame

GRADUATE RESEARCH ASSISTANT

Notre Dame, IN, USA

Aug. 2022 – Present

- **Led a 6-month DoD-funded project**, developing a fully operational **underwater burrowing robot** from concept through fabrication and experimental validation.
- **Designed and optimized bio-inspired soft robotic systems** for terrestrial, underwater, and surface-water exploration, including multi-legged platforms, **sea star-inspired robots**, and **soft grippers**.
- **Performed nonlinear FEA in Abaqus** on fabric-reinforced silicone soft pneumatic actuators, PLA sea-star rays, and silicone artificial skin, analyzing bending, stress, FSI using CEL, and anisotropic friction.
- **Integrated a Robotiq FT 300-S force/torque sensor** with a custom mount to measure forces generated by soft robotic legs; performed data acquisition and analysis.
- **Mentored 13 student researchers**, supporting 13+ poster and oral presentations.

Thermal Engineering Lab, Arkansas State University

GRADUATE & POSTGRADUATE RESEARCH ASSISTANT

Jonesboro, AR, USA

Jan. 2015 – Aug. 2018

- Developed predictive models for ESP and SCR systems in an 870 MW supercritical power plant, modeling NO_x reduction and ash deposition.
- Contributed to \$1.22M federally funded research; modeled VRFB systems (ASTA \$79,379) and supported NSF EPSCoR energy storage test-bed development (\$120,000).

Technical Projects

- **Soft Rigid Hybrid Modular Terrestrial Robot**: Developing a hybrid soft rigid locomotor integrating a plastic skeletal frame, silicone compliant structure, and custom BLDC motor gearbox system for modular terrestrial locomotion.
- **Baja SAE eCVT Mechanical Design Project**: Collaborating with the University of Notre Dame Baja Team on the mechanical design and development of an electronic continuously variable transmission for the Baja SAE vehicle.
- **ANDBot: Reconfigurable Amphibious Robot for Cross Environmental Exploration**: Designed a reconfigurable robotic platform for locomotion across terrestrial and aquatic environments.
- **Shape Adaptive Soft Gripper for Picking Thin and Small Objects**: Designed and developed a compliant soft gripper for grasping small, thin, and difficult to pick objects.
- **Wearable Soft Finger Rehabilitation Robot**: Developed a wearable pneumatic soft actuator system for controlled finger bending and rehabilitation assistance.

Selected Publications

JOURNAL ARTICLES

- **Siddiquee, A. N. M. A., & Ozkan-Aydin, Y.** Design and Simulation of Soft, Multi-Legged Robotic Swimmer for Shallow Water Locomotion. [Under Review]
- **Siddiquee, A. N. M. A., & Ozkan-Aydin, Y.** (2024). Utilizing Bioinspired Soft Modular Appendages for Grasping and Locomotion in Multi-Legged Robots on Ground and Underwater. IEEE Robotics and Automation Letters (RA-L).

CONFERENCE PAPERS

- Siddiquee, A. N. M. A., Castillo Gonzalez, M., Utama, V., Zhu, Y., & Ozkan-Aydin, Y. (2026). Design and Iterative Evaluation of a Sea Star-Inspired Robot with Autonomous Ray. **[Accepted: IEEE IROS 2026]**
- **Siddiquee, A., & Jeong, K.** (2016). Conjugated Dynamic Modeling on Vanadium Redox Flow Battery With Non-Constant Variance for Renewable Power Plant Applications. Proceedings of the ASME 2016 International Mechanical Engineering Congress and Exposition (IMECE2016-67462), Phoenix, AZ, USA.

Professional Leadership & Recognition

- **Edison Innovation Fellow**; Notre Dame IDEA Center (2026–2027) — Selected as 1 of 10 university-wide fellows (\$17,077.50 fellowship stipend.)
- **Peer Reviewer, IEEE IROS 2024**; presented peer-reviewed research at **IEEE ICRA, ASME IMECE**.
- Competitive travel awards: **IEEE ICRA 2025 (\$1,500)**, **James L. Massey (3 × \$750)**, and **A-State (\$1,500)**.